Facial Expression Recognition on FER2013

Controlled experiments with a custom CNN, ResNet18, EfficientNet-B0, and CK+ external validation

Shaoya Zhang and Laura Guitart

CONTENT

Presentation structure

1. Introduction and motivation
2. Related work / state of the art
3. Dataset and preprocessing
4. Methodology and implemented models
5. Experiments and evaluated configurations
6. Results and analysis
7. Comparison against baselines and state-of-the-art approaches
9. Conclusions and future work

The goal of our project: emotion classification

- Identifying facial expressions can be interesting for human-computer interactions, to prevent accidents by drivers falling asleep, blind people being able to detect emotions, etc.

Our goal:

Evaluation of:

- **Architectures**
- **Transfer learning techniques**
- **Training approaches**

→ **For the classification of FER2013 dataset**

Similar researches for FER2013 emotion classification

DTL-I-ResNet18: facial emotion recognition based on deep transfer learning and improved ResNet18

- Comparison of different popular CNNs architectures for FER2013 classification problem.
- They achieved a macro F1 Score of 64%.

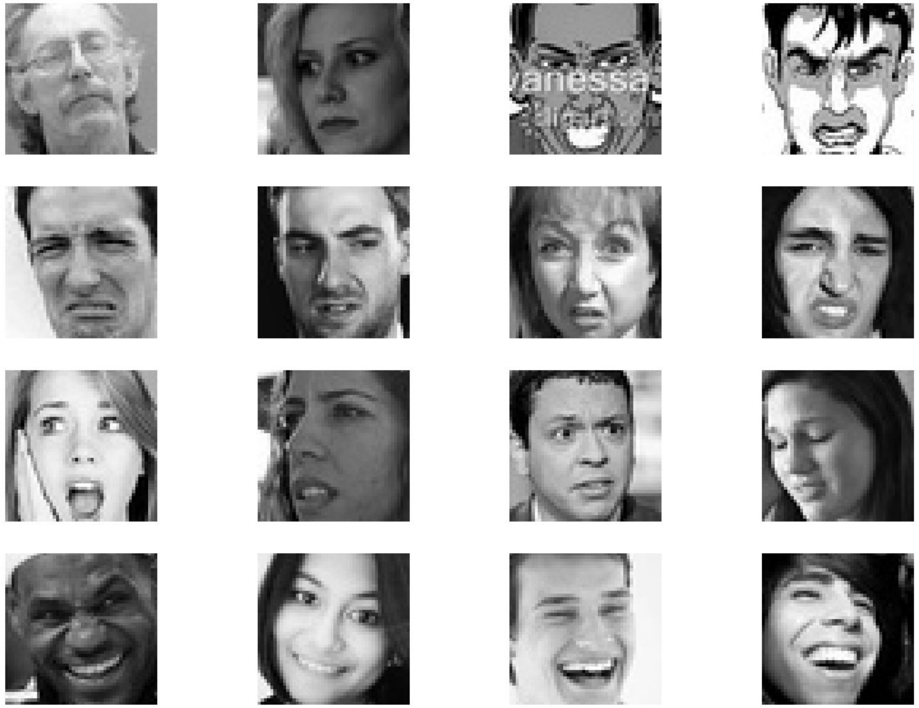
DTL-I-ResNet18: facial emotion recognition based on deep transfer learning and improved ResNet18 | Request PDF

FacialExpression Recognition Using Deep Learning EfficientNetB0

- They compared two Transfer Learning techniques for Efficientnetb0.
- They achieved a macro F1 Score of 72%.

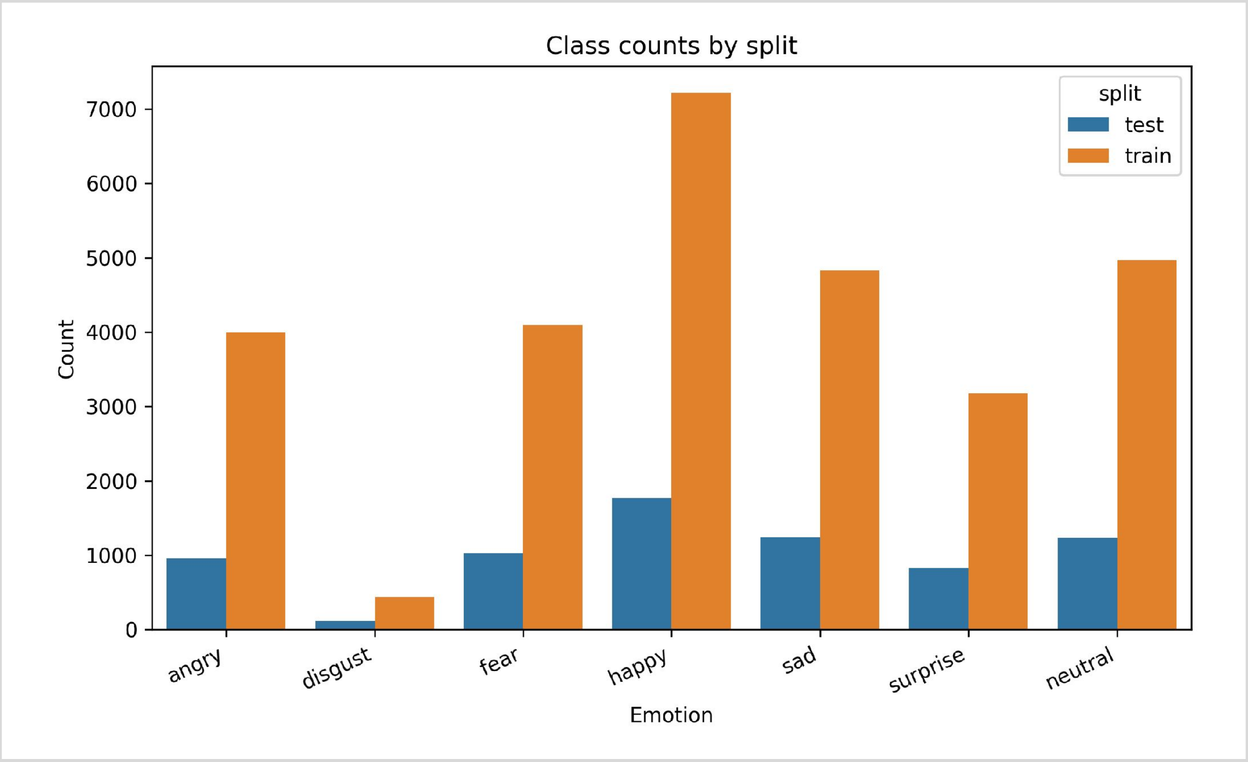
https://www.ijs.uobaghdad.edu.iq/index.php/eijs/article/view/13867/7846

FER2013: Sample images reveal ambiguity and low resolution.



- Faces are small (48x48), grayscale, and sometimes poorly aligned.
- Several classes differ by subtle mouth, eye, or eyebrow cues.
- Human-visible ambiguity justifies evaluating more than a single aggregate score.

Class imbalance and affect proximity explain why this dataset is difficult.



- Happy has much more support than minority classes such as disgust.
- We will need to handle this class imbalance during the training of our models

The split is fixed, and validation is created from the training folder.

FER2013 train — stratified split with fixed seed — 90% train — 10% validation

FER2013 official test — full held-out test set

Dataset size

28,709 train

7,178 test

- The train/validation split is stratified and reproducible with seed 42.
- The official FER2013 test set is kept fixed and used without resampling.

Emotion	Test	Train
Angry	0.13	0.14
Disgust	0.02	0.02
Fear	0.14	0.14
Happy	0.25	0.25
Neutral	0.17	0.17
Sad	0.17	0.17
Surprise	0.12	0.11

Input preprocessing depends on whether the model is trained from scratch or pretrained.

CUSTOM CNN

- 48x48 grayscale input
- Horizontal flip, affine transformations and rotations
- mean 0.5 / std 0.5 normalization

RESNET18 AND EFFICIENTNET-B0

- same transfer preprocessing for both pretrained backbones
- resize to 224x224 and replicate grayscale to 3 channels
- ImageNet normalization to match pretrained weights

The methodology is built around controlled experiments/ablation.

Fix split — Change one factor/ablation — Train / select checkpoint — Evaluate same metrics — Analyze errors

- The experiment compares model families under consistent train/validation/test logic.
- Variables include architecture, imbalance handling, loss function, optimizer, transfer learning techniques, crop/TenCrop evaluation, and depth addition.
- Checkpoint selection is based on validation macro F1 and validation loss as tie breaker

Architecture: custom CNN baseline trained from scratch.

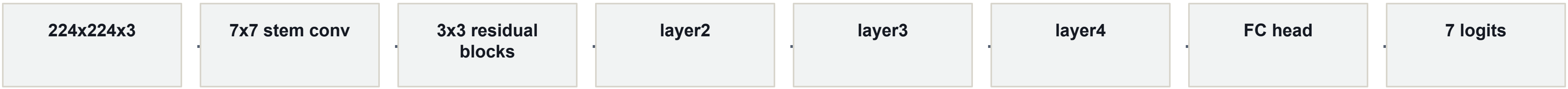
48x48x1 – 3x3 conv 32 x2 – MaxPool – 3x3 conv 64 x2 – MaxPool – 3x3 conv 128 – MaxPool – FC 256 – 7 logits

Baseline CNN implementation

1,321,191 Trainable Parameters

- Feature channels: 32, 64, 128.
- Convolution filters: 3x3, padding 1; block depths: 2, 2, 1.
- Dropout2d after early pooling blocks; dropout before classifier output.
- Purpose: establish a reproducible from-scratch baseline.

Architecture: ResNet18 transfer learning.



~11.69M Trainable Parameters

Why RestNet18 as Pretrained Model

- Standard and widely used transfer-learning baseline for image classification.
- Residual connections make deeper convolutional networks easier to optimize.
- Moderate model size: stronger than a small custom CNN but still practical for FER2013.
- Easy to fine-tune systematically by progressively unfreezing later layers.

Architecture: EfficientNet-B0 transfer learning.

224x224x3 → 3x3 stem conv → MBConv 1-5 → last MBConv blocks → pooling → FC head → 7 logits

5,288,548 parameters

Why include Efficient-B0?

- Lightweight ImageNet-pretrained backbone, much smaller than ResNet18.
- MBConv blocks and depthwise convolutions make feature extraction parameter-efficient.
- Tests whether a compact pretrained model can achieve competitive FER2013 performance.
- Useful comparison against ResNet18: standard transfer backbone vs efficient transfer backbone.

The evaluation uses complementary metrics, not only one score.

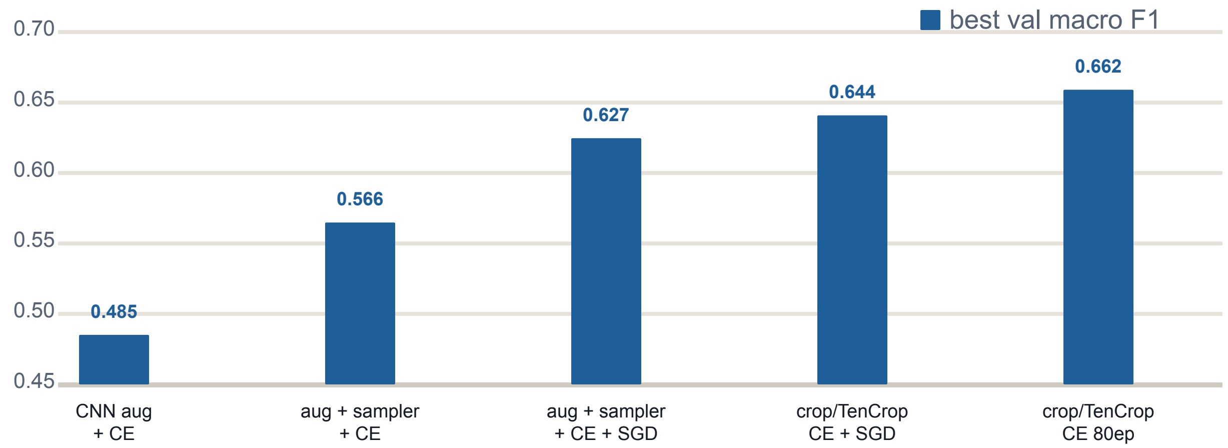
MAIN METRICS

- Macro F1: primary metric because all classes count equally.
- Weighted F1: reflects performance under the real class distribution.
- Precision and recall: show whether each class is over- or under-predicted.

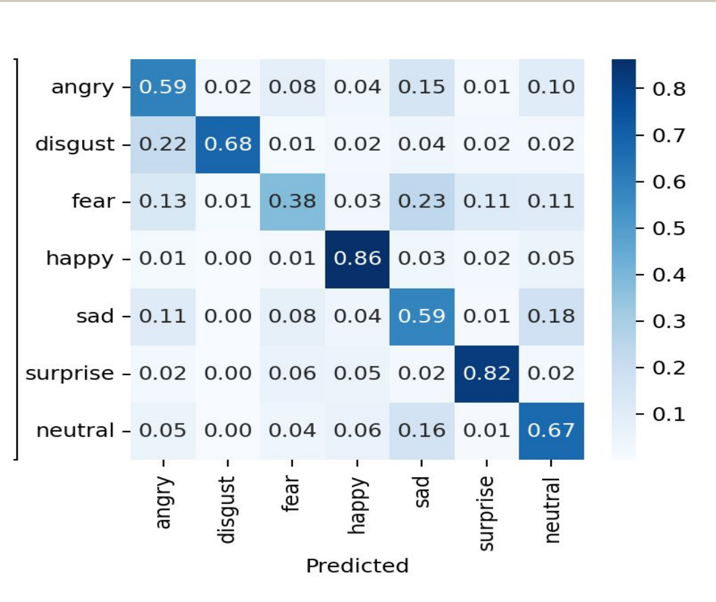
SUPPORTING EVIDENCE

- Loss curves: diagnose optimization and overfitting.
- F1 per class: identifies weak emotion categories.
- Confusion matrix: shows the actual error pattern.

Custom CNN results improve gradually, but plateau below transfer learning.



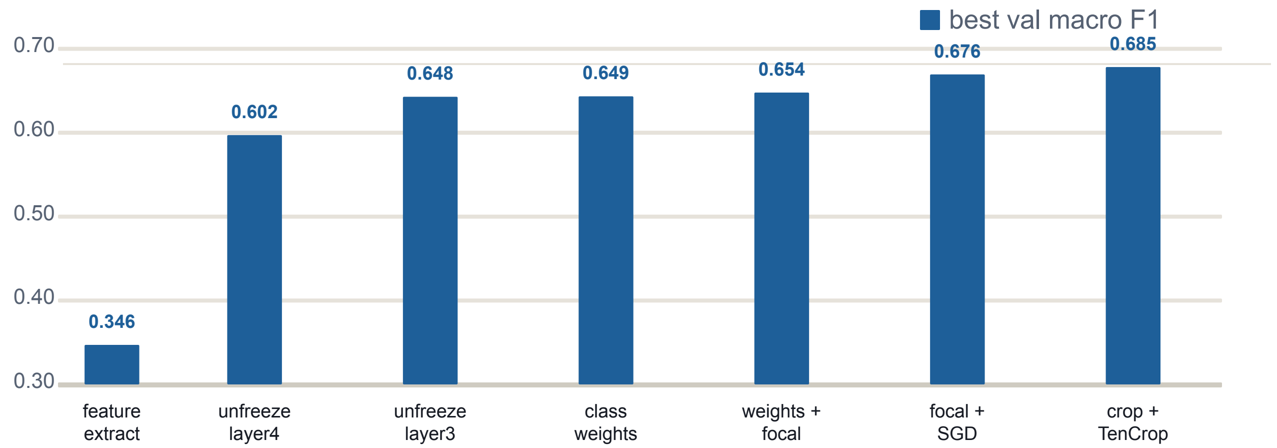
Test Macro F1	0.663	Test Weighted F1	0.675



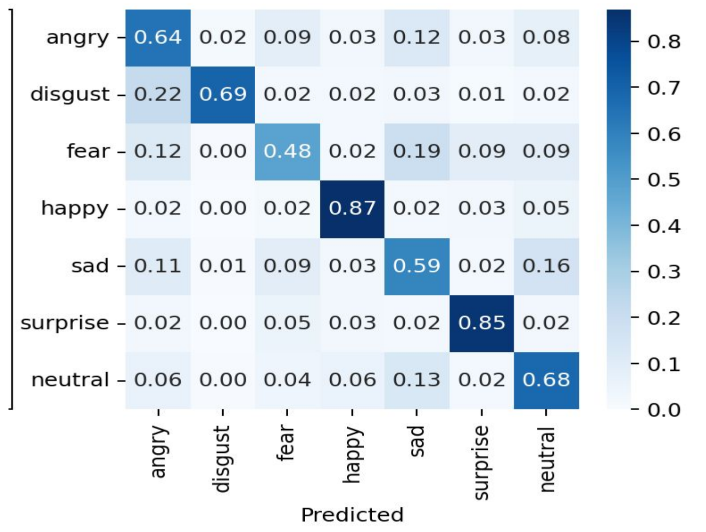
Selected custom CNN test per-class precision / recall / F1

metric	angry	disgust	fear	happy	sad	surprise	neutral
precision	0.59	0.67	0.57	0.87	0.53	0.77	0.61
recall	0.59	0.68	0.38	0.86	0.59	0.82	0.67
F1	0.59	0.68	0.45	0.87	0.55	0.79	0.64

ResNet18 improves when deeper layers are fine-tuned.



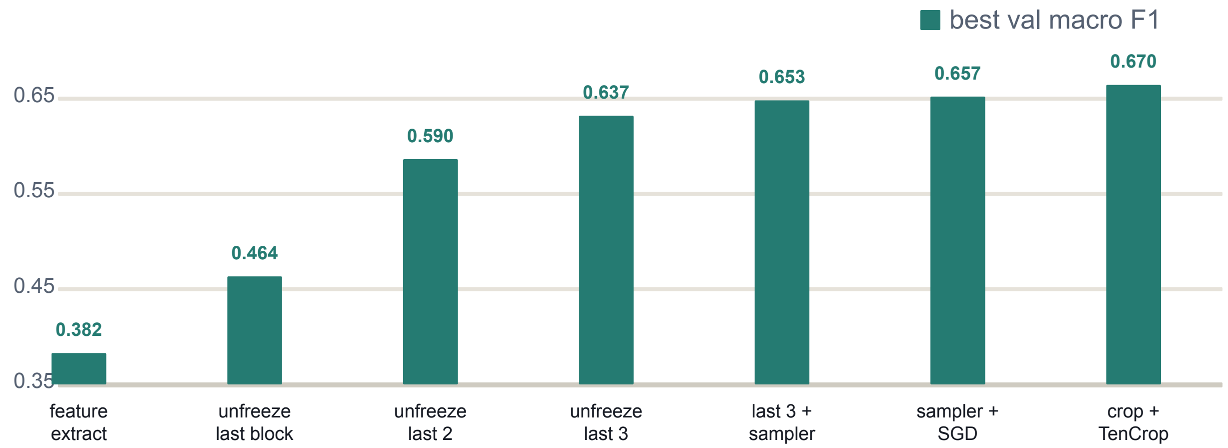
Test Macro F1	0.685	Test Weighted F1	0.695



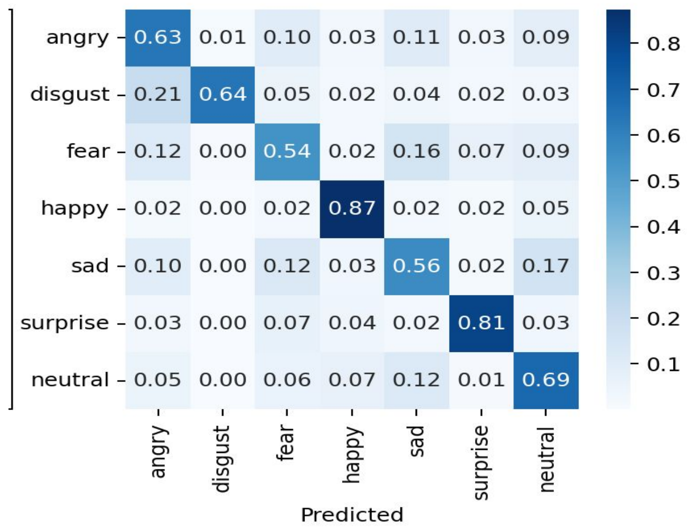
Best ResNet18 test per-class precision / recall / F1

metric	angry	disgust	fear	happy	sad	surprise	neutral
precision	0.60	0.69	0.60	0.90	0.58	0.77	0.64
recall	0.64	0.69	0.48	0.87	0.59	0.85	0.68
F1	0.62	0.69	0.53	0.88	0.58	0.81	0.66

EfficientNet-B0 is competitive on FER2013 after stronger fine-tuning.



Test Macro F1	0.667	Test Weighted F1	0.686



Best EfficientNet-B0 test per-class precision / recall / F1

metric	angry	disgust	fear	happy	sad	surprise	neutral
precision	0.62	0.77	0.56	0.88	0.59	0.81	0.63
recall	0.63	0.64	0.54	0.87	0.56	0.81	0.69
F1	0.62	0.70	0.55	0.88	0.58	0.81	0.65

Top confusions show that errors cluster around similar emotions.

MODEL	TOP CONFUSION	COUNT	CLASS RATE
Custom CNN	sad -> neutral	259	20.8%
Custom CNN	fear -> sad	233	22.8%
Custom CNN	neutral -> sad	162	13.1%
ResNet18	sad -> neutral	200	16.0%
ResNet18	fear -> sad	197	19.2%
ResNet18	neutral -> sad	162	13.1%
EfficientNet-B0	sad -> neutral	206	16.5%
EfficientNet-B0	fear -> sad	163	15.9%
EfficientNet-B0	sad -> fear	154	12.3%

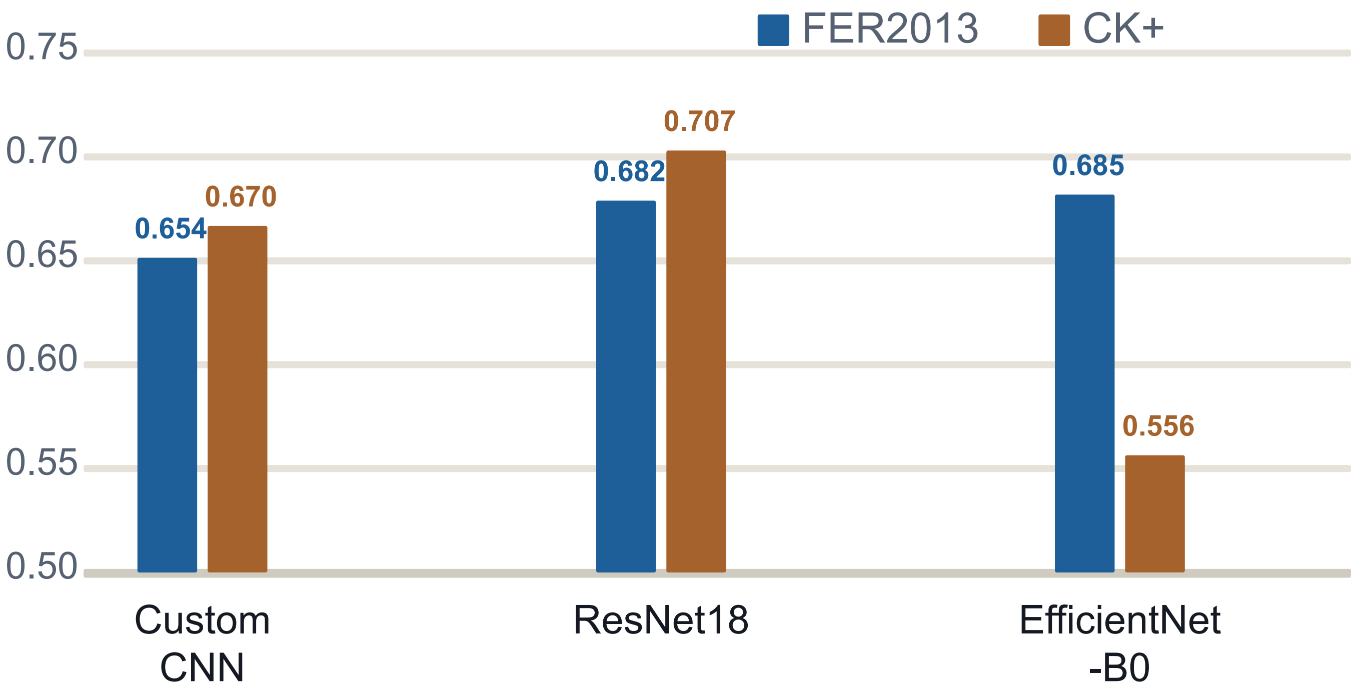
- Sad, fear, and neutral dominate the top mistakes.
- The same error families appear across model architectures.
- The best model improves the score, but does not remove class ambiguity.

Comparison of our models with literature results

MACRO F1 COMPARISON BETWEEN OUR RESULTS AND STATE OF THE ART

	Literature Resnet18	Our Resnet18	Literature EfficientNet-B0	Our EfficientNet-B0	Our Baseline model
Neutral	0.57	0.66	0.70	0.65	0.64
Happy	0.48	0.88	0.91	0.88	0.87
Sad	0.52	0.58	0.58	0.58	0.55
Surprise	0.79	0.81	0.84	0.81	0.79
Fear	0.50	0.53	0.58	0.55	0.45
Disgust	0.68	0.69	0.78	0.70	0.68
Angry	0.57	0.62	0.65	0.62	0.59
Macro-average	**0.64**	**0.69**	**0.72**	**0.67**	**0.66**

FER2013 and CK+ MacroF1 comparisons show promising results for ResNet18 and our custom CNN

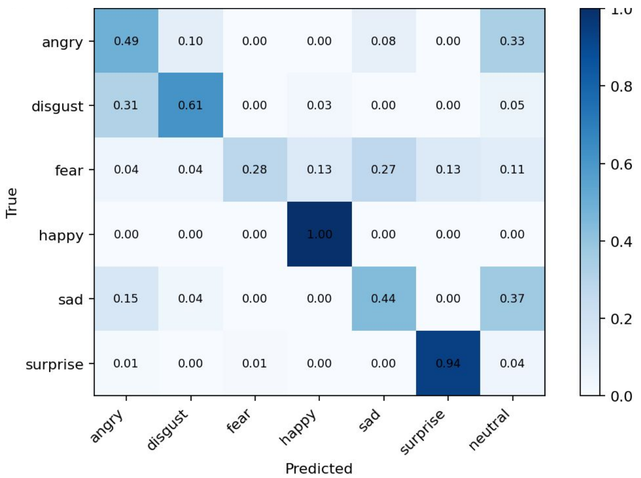


EfficientNet was not able to mantain performance regarding MacroF1.

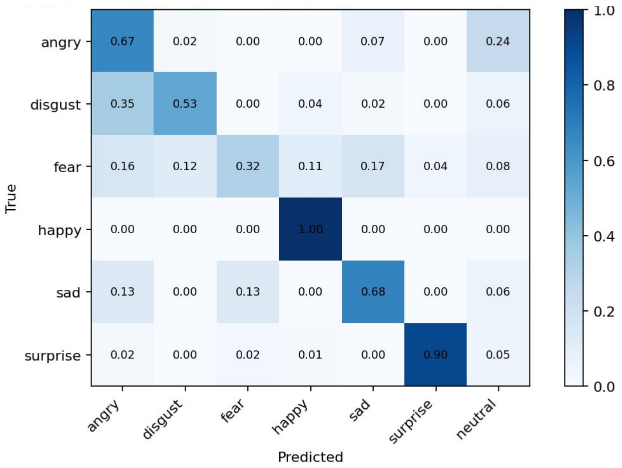
This probably means it was not able to generalize expressions features, and learnt specifically the ones from FER2013.

CK+ performance is worse in most emotions

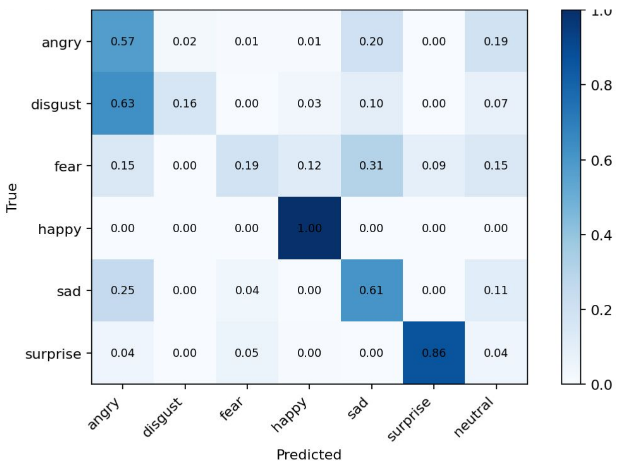
Custom CNN CK+



ResNet18 CK+



EfficientNet-B0 CK+



CK+ Custom CNN per-class F1

angry	disgust	fear	happy	sad	surprise
0.48	0.71	0.43	0.97	0.49	0.95

CK+ ResNet18 per-class F1

angry	disgust	fear	happy	sad	surprise
0.57	0.66	0.41	0.96	0.69	0.94

CK+ EfficientNet-B0 per-class F1

angry	disgust	fear	happy	sad	surprise
0.42	0.28	0.26	0.96	0.51	0.91

The project improves balanced FER2013 recognition while showing where it still fails.

CONCLUSIONS

- Transfer learning provides moderate improvements over the custom CNN baseline, even though they were trained for different tasks
- Not all emotions are equally distinguishable
- Per-class analysis is essential because weak classes remain hidden in aggregate scores
- Testing on external datasets can help to find issues in our models

FUTURE WORK

- Establish human performance baseline
- Clean noisy labels and use soft labels for ambiguous samples
- Extend external validation beyond CK+ and test stronger or ensemble backbones

Thank you for your attention!